A Theory of Economic Slack

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CHAPTER 16.

Monetary policy

We have seen that the unemployment rate in the United States is always inefficiently high in slumps, and sometimes inefficiently low in booms (chapter 12). Therefore, the economy requires active and systematic stabilization policy to align unemployment with its efficient level. A natural policy to achieve such stabilization is monetary policy, which has scope to stabilize the unemployment rate. The link is the interest-rate channel: by shifting interest rates, monetary policy can move aggregate demand and thus unemployment toward the efficient rate—at least when conventional interest-rate policy is unconstrained (chapter 13).

16.1. Social planner's problem

To study optimal monetary policy, we follow the approach we used to study the unemployment gap in chapter 12: we specify a generic framework and solve the social planner's problem in that framework, which we then use to derive a sufficient-statistic formula for optimal monetary policy.

16.1.1. Economic slack

We assume that there is slack in the economy. This means that a share $u \in (0, 1)$ of the aggregate productive capacity cannot be sold. Furthermore, we assume that slack determines social welfare, and so that social welfare is maximized when the slack rate is at its efficient level, u^*. Last, we assume that the efficient slack rate can be measured by sufficient statistics, as in chapter 12. The models of chapters 13 and 14 satisfy these assumptions.

The model of chapter 15 has two types of slack: product market slack and labor market slack. So social welfare generally depends on the two slack rates. But if the product-labor coincidence holds in that model—which requires that the real wage satisfies condition (15.22)—then product market slack and labor market slack are efficient at the same time. Thus, monetary policy can maximize social welfare by simply maximizing welfare on one of the markets—by targeting one of the slack rates. Hence, the model also satisfies the assumption.

16.1.2. Inflation

A corollary of the above assumption about slack is that inflation does not independently affect social welfare. This could be because inflation is fixed, as in the models of chapter 13 and 15, or because the inflation-slack coincidence holds, as in the model of chapter 14. This means that either inflation does not respond to monetary policy, or inflation is on target when the slack rate is efficient. This is distinct from saying that inflation has no welfare cost: it is saying that stabilizing slack at its efficient rate does not conflict with price stability.

16.1.3. Monetary policy

In the efficiency analysis of chapter 9, we simply assumed that the planner could pick the amount of slack in the market to maximize welfare. Here, we are studying optimal monetary policy, which means that our planner is more restricted: they can only affect slack indirectly, through monetary policy.

Specifically, we assume that monetary policy is non-neutral and conducted through the nominal interest rate. This means that the planner determines the nominal interest rate i, and the nominal interest rates can stabilize the economy. The slackish models in part IV are an example of this: in those models interest rates control aggregate demand and then output and slack. Formally, the nominal interest rate affects the slack rate through a generic function $u(i)$.

16.1.4. Planner's problem

We now formalize the social planner's problem. The planner uses the nominal interest rate i to maximize social welfare through the slack rate $u(i)$. Because social welfare is maximized at the efficient slack rate u^*, the optimal nominal interest rate i^* satisfies:

$$u(i^*) = u^*. \tag{16.1}$$

16.2. Sufficient-statistic formula for optimal monetary policy

We now derive our sufficient-statistic formula for optimal monetary policy.

16.2.1. Sufficient-statistic formula

Given the current slack rate, u, and current nominal interest rate, i, the sufficient-statistic formula gives the optimal nominal interest rate, i^*, that the central bank should set.

To derive the formula, we perform a first-order Taylor expansion of the function $u(i)$ around i:

$$u(i^*) = u(i) + \frac{du}{di} \cdot \left(i^* - i\right),$$

where the derivative du/di is evaluated at the current interest rate and slack rate. By definition of the function $u(i)$ and optimal interest rate i^*, we have $u(i) = u$ and $u(i^*) = u^*$, so the Taylor expansion gives

$$u^* = u + \frac{du}{di} \cdot \left(i^* - i\right).$$

Rewriting this to express the optimal interest rate as a function of the prevailing slack gap $u - u^*$, we get

$$i^* = i - \frac{u - u^*}{du/di}. \tag{16.2}$$

This is our sufficient-statistic formula. The statistic $u - u^*$ is the slack gap. The statistic du/di is the monetary multiplier: it gives the percentage-point change in slack when the nominal interest rate increases by 1 percentage point. It is positive if an exogenous increase in nominal interest rate raises slack, as it typically does in the real world.

16.2.2. Interpretation of the formula

The optimal monetary-policy formula (16.2) tells us that the central bank should adjust the nominal interest rate to bring the slack rate to its efficient level.

Consider the realistic case where the monetary multiplier is positive, so an increase in interest rates raises the slack rate. Then if the slack rate is inefficiently high, the central bank needs to cut interest rates. On the other hand, if the slack rate is inefficiently low, interest rates must be raised.

We can also see that how much the central bank should respond to the slack gap depends on the monetary multiplier. If the monetary multiplier is large, monetary policy has a strong effect on the slack gap so rates need to be changed by a smaller amount. If the multiplier is small, monetary policy only has a limited impact on the slack gap so rates need to be changed by a larger amount.

16.2.3. Calibration of the sufficient statistics

To calibrate formula (16.2), we need empirical measures of the slack rate, the efficient slack rate, and the monetary multiplier. Since aggregate slack is not readily measurable in the United States, we use the unemployment rate as our measure of slack and the FERU as the corresponding efficient slack rate. Accordingly, we measure the monetary multiplier by the response of unemployment to the nominal interest rate.

The nominal interest rate i in the United States is just the federal funds rate controlled by the Board of Governors of the Federal Reserve System (2025).

The US unemployment gap $u - u^*$ can be estimated directly from the unemployment and vacancy rates, as we saw in chapter 12.

A challenge in measuring the monetary multiplier du/di—and measuring the impact of policies in general—is that nominal interest rate changes are never exogenous: they are endogenous to the current situation. This makes it difficult to estimate the monetary multiplier properly. But several methods have been developed to isolate exogenous variations in the nominal interest rate in the United States and measure the effect of monetary policy on aggregate variables, particularly on the unemployment rate.[1]

Many papers use vector autoregressions (VARs) to identify exogenous variations in monetary policy and compute the monetary multiplier. For instance, using such an approach, Coibion (2012, p. 5) finds $du/di = 0.16$. This value accords with previous estimates of the monetary multiplier obtained by Christiano, Eichenbaum, and Evans (1996), Bernanke, Boivin, and Eliasz (2005), and others, with the exception of the larger estimate $du/di = 0.6$ obtained by Bernanke and Blinder (1992).

However, VAR does come with its issues—it is quite difficult to estimate the exogenous changes in monetary policy. An alternative approach is the narrative approach pioneered by Romer and Romer (1989, 2004). The idea is to carefully read accounts of policy decisions to separate endogenous and exogenous changes in the interest rate by isolating idiosyncratic decisions made by the Federal Reserve that were not driven by current economic conditions. Romer and Romer (1989, 2004) construct a series of monetary policy shocks using only these idiosyncratic decisions and, based on this series, measure the impact of monetary policy on the economy. Using the narrative approach, Coibion (2012, p. 7) obtains a much larger estimate of the monetary multiplier: $du/di = 0.93$.

This is significantly bigger than the value computed using VAR, and using this number would lead to very different outcomes in our analysis. Thus, it is important to try to get a narrower range of estimates for the monetary multiplier so that we can be more certain about how to conduct monetary policy. To reconcile the difference between the two

[1] See Christiano, Eichenbaum, and Evans (1999) and Ramey (2016) for surveys.

approaches, Coibion (2012, p. 8) proposes a hybrid approach that yields a medium estimate of the monetary multiplier: $du/di = 0.40$. The hybrid approach appears robust: across numerous specifications, it yields monetary multipliers between 0.4 and 0.6 (Coibion 2012, p. 12). We use the midpoint of this range, $du/di = 0.5$, to carry out our analysis and study optimal monetary policy.

16.3. Optimal response to unemployment fluctuations

Now that we have estimates for the sufficient statistics in our optimal interest-rate formula, we can put everything together to describe what the Fed's optimal behavior is. Combining the optimal interest-rate formula, given by (16.2), with the midrange estimate of the monetary multiplier, $du/di = 0.5$, we obtain the following formula:

$$i^* = i - 2 \times \left(u - u^*\right). \tag{16.3}$$

Therefore, for any unemployment gap, the optimal response of the nominal interest rate is twice the size of the gap. For example, if the unemployment gap is 1pp, the federal funds rate should drop by 2pp.

This is a very simple rule that should guide the Fed's response to unemployment fluctuations. Of course, it differs from the typical Taylor (1993) rule that appears in New Keynesian macroeconomics: rule (16.3) links monetary policy to the unemployment rate whereas the Taylor rule links it to the inflation rate. Moreover, the optimal response of monetary policy to the unemployment rate can be obtained empirically, by estimating the monetary multiplier. By contrast, the most appropriate response of monetary policy to inflation is typically determined by simulations in New Keynesian models.[2]

16.4. Optimal monetary policy without inflation-slack coincidence

The same logic applies to optimal monetary policy in models without inflation-slack coincidence. When the coincidence fails, monetary policy faces a tradeoff between closing the slack gap and bringing inflation to its target, so targeting the efficient slack rate is no longer optimal. In that case, the optimal interest rate is determined by weighting the slack gap against the inflation gap.

To see this, consider an extension of the social planner's framework in which social welfare depends not only on slack but also on inflation—for instance as in chapter 14. Let's denote the efficient inflation rate by π^*. The welfare loss around the efficient allocation

[2] See for instance the survey by Taylor and Williams (2010).

(u^*, π^*) admits the following quadratic approximation:

(16.4) $$\mathcal{L}(u, \pi) = \left(\pi - \pi^*\right)^2 + \Lambda\left(u - u^*\right)^2,$$

where $\Lambda > 0$ measures the importance of slack relative to inflation in the social welfare function.

Additionally, the social planner faces a Phillips curve that relates inflation to slack. Around the efficient allocation, the Phillips curve admits the following linear approximation:

(16.5) $$\pi - \pi^* = -\Gamma\left(u - u^*\right) + \Omega,$$

where $\Gamma > 0$ gives the slope of the downward-sloping Phillips curve, and $\Omega \neq 0$ is introduced to break the inflation-slack coincidence. Indeed, if $\Omega = 0$, then $\pi = \pi^*$ when $u = u^*$: inflation is on target when slack is efficient, so the inflation-slack coincidence holds. But if $\Omega > 0$, then $\pi > \pi^*$ when $u = u^*$: inflation is too high when slack is efficient, so slack must be inefficiently high to bring inflation to target, just like in the 1970s. Conversely, if $\Omega < 0$, then $\pi < \pi^*$ when $u = u^*$: inflation is too low when slack is efficient.

The Phillips curve developed in chapter 14 links the slack and inflation gaps. It guarantees that the inflation-slack coincidence holds if inflation expectations are anchored to the inflation target, so $\Omega = 0$ in (14.20). But the coincidence fails if expectations about normal inflation are not anchored to the inflation target. In that case, the Phillips curve features a shifter $\Omega \neq 0$, as seen in (14.24). The shifter is positive if the inflation that people consider normal is above target ($\pi^n > \pi^*$); the shifter is negative if the inflation that people consider normal is below target ($\pi^n < \pi^*$).

The social planner minimizes the welfare loss (16.4) subject to the Phillips curve (16.5). Inflation can be substituted out of the welfare loss using the Phillips curve (result A.22). Then, the planner's problem is to find $u \in [0, 1]$ to minimize

$$\left[-\Gamma\left(u - u^*\right) + \Omega\right]^2 + \Lambda\left[u - u^*\right]^2.$$

This objective function is strictly convex in u, so the first-order condition is sufficient to find its minimum (result A.19). We take the function's derivative with respect to u and set it to 0:

$$0 = -2\Gamma\left[-\Gamma\left(u - u^*\right) + \Omega\right] + 2\Lambda\left[u - u^*\right].$$

Rearranging terms, we express the optimal slack gap as a function of the parameters:

(16.6) $$u - u^* = \frac{\Omega\Gamma}{\Gamma^2 + \Lambda}.$$

Plugging (16.6) in the Phillips curve (16.5), we express the optimal inflation gap as a function of the same parameters:

$$\pi - \pi^* = \frac{\Omega\Lambda}{\Gamma^2 + \Lambda}. \tag{16.7}$$

From these two expressions, we obtain a simple expression for the ratio between the optimal slack and inflation gaps:

$$\frac{u - u^*}{\pi - \pi^*} = \frac{\Gamma}{\Lambda} > 0. \tag{16.8}$$

The tradeoff between inflation and slack in the absence of inflation-slack coincidence clearly appears in these equations. Consider the case where inflation is too high when slack is efficient ($\Omega > 0$). Then (16.6) and (16.7) say that it is optimal to keep slack above its efficient level ($u > u^*$) and inflation above its target ($\pi > \pi^*$). Furthermore, (16.8) shows that the optimal gaps are determined by the welfare cost of slack relative to inflation (Λ) and the response of inflation to slack (Γ). If slack is particularly costly (high Λ) or if the Phillips curve is flat (low Γ), the optimal slack gap is small relative to the optimal inflation gap. In such cases, the optimal slack rate remains close to the efficient slack rate, either due to the high marginal cost of slack (high Λ) or the limited marginal benefit of slack (low Γ).

The equations provide two additional insights. First, if the inflation-slack coincidence holds ($\Omega = 0$), then it is optimal to keep slack at its efficient level ($u = u^*$ in (16.6)), which guarantees that inflation hits its target ($\pi = \pi^*$ in (16.7)). Second, it is never optimal for the slack and inflation gaps to have opposite signs ($(u - u^*)/(\pi - \pi^*) < 0$ in (16.8)). This is because welfare can be improved if the gaps take opposite signs. For example if the slack gap is negative and the inflation gap is positive (as in 2021–2024), then raising slack moves the slack gap toward 0 and, by cooling inflation, moves the inflation gap toward 0—thereby improving welfare.

16.5. Might the price mandate explain departures from full employment?

Despite the US government's full-employment mandate, the US labor market has consistently fallen short of full employment in the past century (chapter 12). In a world of fixed inflation or inflation-slack coincidence, there is no justification for falling short of full employment. But if the inflation-slack coincidence does not hold, one possible justification is that the dual mandate forced the Federal Reserve to keep unemployment inefficiently high so as to tame inflation, along the lines of equation (16.8). We explore this possibility here.

A central insight of equation (16.8) is that when monetary policy is conducted optimally, the unemployment gap and inflation gap should have the same sign. In other words, an unemployment rate above the FERU can only be justified if inflation is above target—as a way to cool inflation. Conversely, an unemployment rate below the FERU can only be justified if inflation is below target—as a way to boost inflation. In fact, optimality requires that the unemployment gap and inflation gap are linearly related through equation (16.8). A weaker but useful diagnostic is that they must at least have the same sign: if they do not, welfare can be improved by moving them toward the ratio in (16.8).

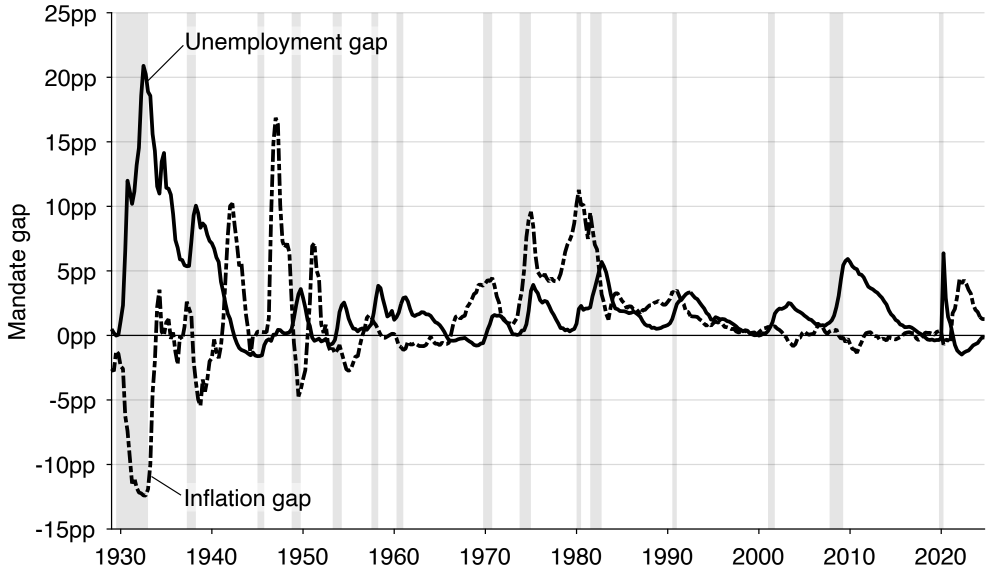
To assess whether this basic optimality requirement is satisfied, we first plot the unemployment gap (the distance between the unemployment rate and FERU) and inflation gap (the distance between the inflation rate and target of 2%) over time (figure 16.1A). Whenever the two gaps have opposite signs, monetary policy cannot be optimal. What we first see is that in many notable recessions, when the unemployment gap is positive, the inflation gap is negative, so clearly monetary policy was suboptimal. There was for instance notable deflation during the Great Depression—when the inflation gap was below −10pp. There were also notable negative inflation gaps, together with positive unemployment gaps, during the 1937–1938, 1948–1949, and 1953–1954 recessions.

To evaluate more systematically monetary policy, and assess whether the price-stability mandate could really explain departures from full employment, we produce scatter plots of the unemployment gap against the inflation gap (figure 16.1B). Under our baseline necessary condition for optimality, the unemployment and inflation gaps must have the same sign (equation (16.8)). Quarters that fall in the northwest or southeast quadrants therefore violate this condition: in those quadrants the gaps have opposite signs, so monetary policy cannot be optimal.

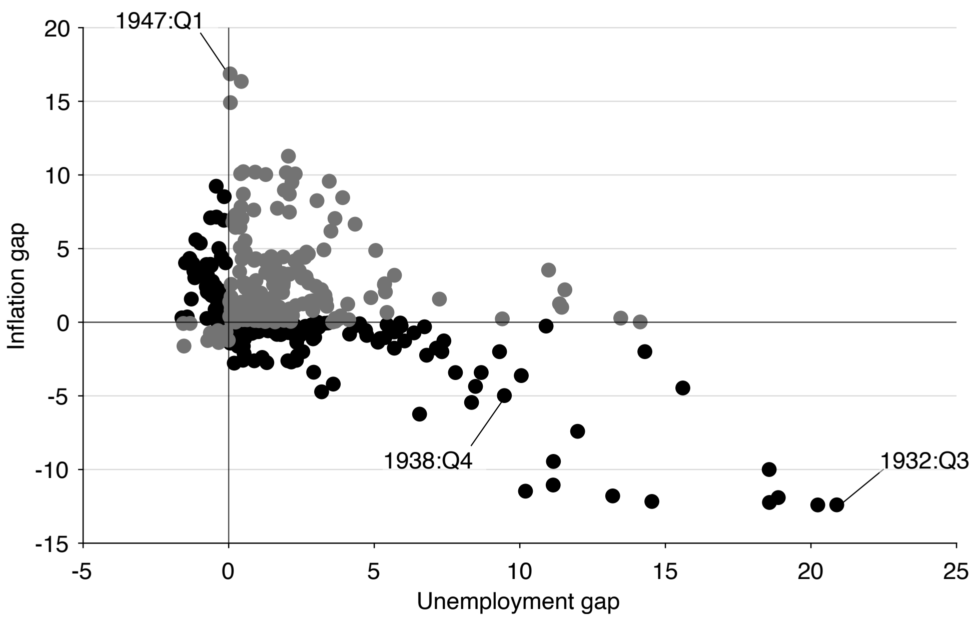
In the entire, 1929–2024 period, monetary policy is non-optimal at least half of the time: 47% of the 384 quarters cannot be optimal. This is not to say that monetary policy is optimal in 53% of the quarters: just that optimality cannot be rejected on the grounds that unemployment and inflation gaps have opposite signs. Indeed, when monetary policy is conducted optimally, the unemployment and inflation gaps should be linearly related (equation (16.8)). Such a linear relationship does not appear in the scatter plots.

One possible issue with the test of monetary policy in figure 16.1 is that the Federal Reserve did not officially adopt a 2% target for inflation until January 2012 (Board of Governors of the Federal Reserve System 2012). However, inflation targeting spread rapidly around the globe in the 1990s, with many countries adopting an inflation target of 2% (Svensson 2010, table 1). Even in the United States, Federal Reserve officials started openly discussing numerical targets for inflation in the 2000s, often around 2% (Shapiro and Wilson 2019).

So we now examine whether our test delivers different results in the modern era, 2000–2024, once the 2% inflation target was more or less in place. In figure 16.2A, we zoom



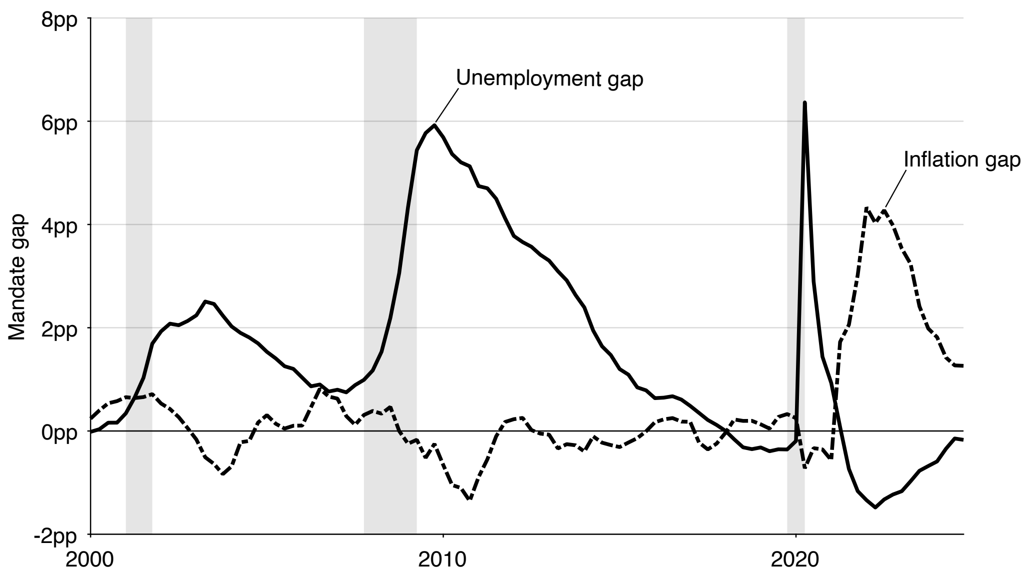
A. Time-series representation



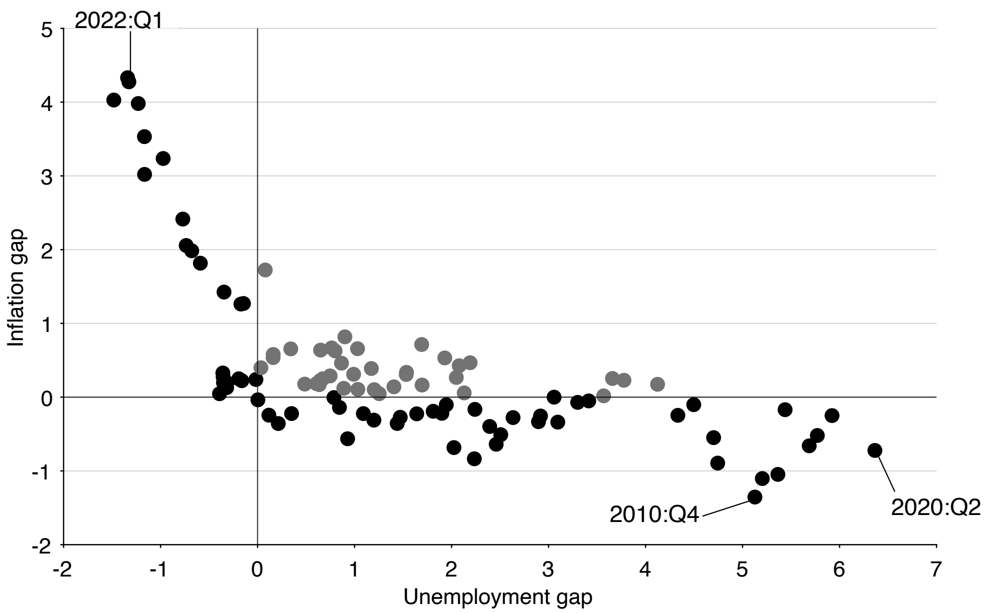
B. Scatter-plot representation

FIGURE 16.1. Unemployment gap and inflation gap in the United States, 1929–2024

The unemployment gap is $u - u^*$, where the unemployment rate u comes from figure 2.1 and the FERU u^* comes from figure 12.2. The inflation gap is $\pi - 2\%$, where π is the quarterly average of the monthly, seasonally adjusted, percentage change from a year ago of the consumer price index for all urban consumers which is produced by the BLS (2025a). After 1958, we use the core consumer price index produced by the BLS (2025b), which is typically preferred by the Federal Reserve because it does not include the prices of food and energy. Shaded areas indicate recessions dated by the NBER (2023).



A. Time-series representation



B. Scatter-plot representation

FIGURE 16.2. Unemployment gap and inflation gap in the United States, 2000–2024

The unemployment gap is $u - u^*$, where the unemployment rate u comes from figure 2.1 and the FERU u^* comes from figure 12.2. The inflation gap is $\pi - 2\%$, where π is the quarterly average of the monthly, seasonally adjusted, percentage change from a year ago of the consumer price index for all urban consumers, less food and energy, which is produced by the BLS (2025b). Shaded areas indicate recessions dated by the NBER (2023).

in on the 2000–2024 period. We see again that the large unemployment gaps observed after the dot-com recession (+2.5pp), and especially the Great Recession (+5.9pp) and pandemic recession (+6.3pp) cannot be explained by the price-stability mandate. When the unemployment gap peaked after these recessions, the inflation gap was always negative. This is especially true after the Great Recession, when the inflation gap fell as low as −1.4pp; the inflation gap bottomed at −0.8pp after the dot-com recession and −0.7pp after the pandemic recession.

Another striking period of suboptimal monetary policy is the post-pandemic period. The inflation gap was positive from 2021:Q2 until the end of 2024, peaking at +4.3pp in 2022:Q3, but that cannot be justified by a positive unemployment gap. To the contrary, unemployment was below the FERU from 2021:Q3 onward, and the unemployment gap fell as low as −1.5pp in 2022:Q2. So in 2021–2024, monetary policy was unambiguously too lax, leading to excessive inflation and labor market tightness.

Because the idea of targeting an inflation rate of 2% was not widespread until 2000, it is possible that monetary policy does better in the scatter-plot test in the 2000–2024 period. One might expect that the share of non-optimal quarters is lower after 2000, once the concept of price stability aligns with the 2% used in the test. That is not what we observe, however. Over the 2000–2024 period, we see that 63% of the 100 quarters display non-optimal monetary policy (figure 16.2B).

We finally zoom in on the even more recent period, 2008–2024, when the inflation target of 2% was generally accepted at the Federal Reserve (Shapiro and Wilson 2019, figure 1). We find that 81% of the 68 quarters display non-optimal monetary policy! That is, since the Great Recession, only 19% of the quarters feature same-sign unemployment and inflation gaps. Clearly, modern departures from full employment are not because of the price-stability mandate.

For the most part, therefore, we can reject the explanation that the US government has failed to maintain the labor market at full employment over the past century because of its price-stability mandate. The only period when large departures from full employment could be explained by the price-stability mandate is the 1970s and 1980s. During these two decades, unemployment was above the FERU and inflation was above target, so any reduction in unemployment might have been deemed undesirable as it might have pushed inflation even higher. Maybe surprisingly, the stagflation of the 1970s is a sign that the Fed was doing the best it could, given the constraints imposed by the Beveridge and Phillips curves. In that way, if the Fed had not let unemployment rise during the period, it would not have done enough to combat rampant inflation.

16.6. So why did the US economy fall short of full employment for a century?

The US government has consistently failed to maintain the labor market at full employment over the past century, and this failure cannot easily be explained by the price-stability mandate. Here we review other explanations as to why the US labor market has deviated from full employment in different periods.

16.6.1. Great Depression and its aftermath

During the Great Depression and its aftermath, the US economy was exceedingly slack. From the beginning of 1930, when our data begin, to the end of 1941, when the United States entered World War 2, the unemployment gap averaged +9.6pp. So the US economy was extremely far from full employment.

Three factors may explain this large amount of slack. The first is that the US government and the Federal Reserve did not have a full-employment mandate at the time. The mandate was introduced with the Employment Act of 1946, as a result of the Great Depression. A second factor is that the Federal Reserve was committed to the gold standard. The gold standard generated a deep deflation in the early 1930s, with dramatic consequences (Eichengreen and Temin 2000). A third factor is that the Fed failed to curb recurrent banking panics in the 1930s (Friedman and Schwartz 1963, chapter 7). Overall, as former Fed Chair Bernanke (2022, p. xvii) writes, "Blaming the Depression entirely on the Fed is an exaggeration, but the relatively new and unseasoned central bank did perform poorly."

16.6.2. World War 2, Korean War, Vietnam War

The US labor market was pulled out of its Great Depression slackness by World War 2 (figure 12.1). In fact, the labor market became inefficiently tight during the war, with tightness averaging 2.8 over the 1942–1945 period. The labor market was once again inefficiently tight during the Korean War, with tightness averaging 1.12 over the 1951–1953 period, and during the Vietnam War, with tightness averaging 1.22 over the 1966–1969 period.

Why was tightness so high during the wars? Part of the reason is that government spending was substantial during these three major wars (Ramey and Shapiro 1998). Such expenditure boosts aggregate demand, which increases tightness. Another part of the reason is that millions of potential labor-force participants are recruited by the military (Department of Veteran Affairs 2023). Such drastic reduction in civilian labor force reduces labor supply, which raises tightness and reduces the unemployment rate among civilian workers.

So why didn't the Fed tighten monetary policy to reduce tightness in wartime? Indeed, a high real interest rate curbs aggregate demand, which reduces tightness and raises unemployment. An appropriate increase in interest rates could have brought tightness back to its full-employment level of 1. In the case of World War 2, there is a simple answer. As Bernanke (2022, p. xviii) explains, during and shortly after World War 2, "at the Treasury's request, the Fed held interest rates at low levels to reduce the government's cost of financing the war."

The same happened at the beginning of the Korean War, when "facing new hostilities in Korea, President Truman pressed the Fed to keep rates low" (Bernanke 2022, p. xviii). The Fed did rebel and was allowed to phase out the low interest-rate peg that had been in place. But the phasing out came too late to cool down the labor market.

The situation during the Vietnam War was different (Bernanke 2022, pp. 20–22). The Fed raised interest rates by half a percentage point at the end of 1965, at the exact time when the economy had reached full employment. However, President Johnson was furious that the Fed tightened monetary policy. He needed low rates to help finance the war. Despite the pressure exerted by Johnson, the Fed continued increasing rates in 1966, which rapidly cooled the labor market. Worried about a possible recession, the Fed reversed its previous tightening. Under pressure from the White House, and facing a chaotic political situation, the Fed continued swinging between tightening and loosening until 1970. The lack of decisive tightening explains why the labor market remained so hot from 1966 to 1969.

16.6.3. Postwar period

In the postwar period, the US labor market was generally inefficiently slack. A manifestation of such pervasive slack is that the unemployment gap averaged 1.4pp between 1946 and 2019. Another manifestation is that the labor market did not achieve full employment once between 1970:Q1 and 2018:Q1—it was inefficiently slack for almost half a century.

A reason that might explain the slackness of the US labor market in the postwar period, especially after 1970, is that the Fed prioritized inflation at the expense of unemployment. Thornton (2011) reviews policy directives by the Federal Open Market Committee (FOMC) and finds that it made no reference to unemployment or full employment between 1979 and 2008—despite the dual mandate introduced in 1977. Instead, Thornton finds that the FOMC preferred "to state its objectives in terms of price stability and economic growth." This changed at the end of 2008, when the FOMC started mentioning its dual objective of "maximum employment and price stability" in policy directives and statements. Kaya et al. (2019) also detect this focus on inflation in FOMC transcripts. They find that from 1960 to 2010 FOMC discussions increasingly emphasized inflation relative to unemployment, and that this shift occurred during the Volcker era and continued even as inflation declined. They conclude that "the emphasis on inflation has become entrenched and disconnected from actual inflation."

The prioritization of inflation might be due to a change in the Fed's preferences or in macroeconomic theory. It might also come from Congress. Hess and Shelton (2016) examine legislative activity to determine when Congress pressures the Fed, and whether this pressure affects monetary policy. They find that by the late 1980s Congress shifted from threatening the Fed when unemployment was high to threatening when inflation was high. This finding is consistent with Weir (1987, p. 377)'s view that "By the mid-1980s full employment had been all but erased as a major political issue in the United States." In fact, Weir (1987, p. 395) argues that although the Kennedy CEA identified an unemployment rate of 4% as full employment, in the following decades "more conservative economists [offered] ever-increasing rates of unemployment as the 'true' definition of full employment."

16.6.4. Great Recession

The Great Recession saw the highest unemployment gap of the 1946–2019 period, at +5.9pp, and it presented new challenges to the Fed. Although the unemployment gap skyrocketed in 2008–2009, the Fed was unable to respond because it ran against the zero lower bound on nominal interest rates from the end of 2008 until 2015 (Board of Governors of the Federal Reserve System 2025). The Fed could not stimulate aggregate demand through lower interest rates because it was constrained by the zero lower bound, so it could not boost tightness and lower unemployment. Hence, unemployment remained inefficiently high until 2018.

Indeed, during the Great Recession, the unemployment gap shot up to around 6 percentage points. In this case, using our formula, the Fed should have lowered the federal funds rate by 12 percentage points. In reality, the federal funds rate at that time was 5%, so it would not have been possible for it to drop so much due to the zero-lower-bound (ZLB) on the nominal interest rate. The ZLB therefore became binding.

The Fed did resort to unconventional monetary policy, including forward guidance and quantitative easing, to reduce long-term interest rates in the aftermath of the Great Recession (Kuttner 2018). But the effectiveness of such policies is debatable (Greenlaw et al. 2018; Michaillat and Saez 2021). Moreover, the Fed may not have used these policies aggressively enough because once again they targeted an unemployment rate that was too high. The Fed commonly uses the CBO's NRU to indicate full employment. During the Great Recession the CBO (2021) adjusted the NRU upward by 1pp because they believed that structural factors temporarily kept the unemployment rate high. As a result, in 2011:Q4, the short-term NRU reached 5.8%. We do find that the outward shift of the Beveridge curve after the Great Recession raised the FERU by 0.5pp, but the FERU only stood at 4.5% in 2011:Q4—1.3pp below the short-term NRU.

16.6.5. Coronavirus pandemic

The coronavirus pandemic sharply slowed down economic activity. In 2020, the US economy reached the largest unemployment gap since the Great Depression, at +6.3pp. As during the Great Recession, the Fed could not respond more aggressively to the slackness of the economy because of the zero lower bound (Board of Governors of the Federal Reserve System 2025).

Thanks to aggressive expansionary fiscal policy, however, the US economy recovered rapidly from the pandemic (Romer 2021). The US economy reached full employment in 2021:Q2, and the labor market continued tightening after that. In 2022:Q2, labor market tightness reached 1.98, a level it had not seen since the end of World War 2. It is only then, in the spring of 2022, that the Fed started tightening monetary policy.

It is unclear why the Fed did not start tightening monetary policy earlier. After the labor market reached full employment (2021:Q2), an entire year passed before the Fed increased rates (2022:Q2). This delay is all the more surprising since inflation was also above its target of 2% at the time. Core inflation was 3.7% in 2021:Q2 and rose to 6.3% in 2022:Q1 (figure 16.2). Combined with the two years required by monetary policy to be fully effective (Coibion 2012), this delay explains well why the labor market remained inefficiently tight until the end of 2024.

One possible explanation is that the Fed relied on the unemployment rate alone to judge whether the economy had recovered to full employment. It is true that the unemployment rate fell to the FERU in 2021:Q2, but the FERU was elevated at that time, at $u^* = 5.8\%$ (figure 12.2). This is because the Beveridge curve shifted far out during the pandemic, which raised the FERU (figure 2.7). So if the Fed was not aware of the increase in the FERU after the pandemic, they would see an unemployment rate that was much higher than before the pandemic, and would conclude that the economy had not reached full employment yet. Indeed, in the summer of 2021, Powell (2021, p. 4), the chairman of the Fed, offered the following assessment of the labor market:

> The unemployment rate has declined to 5.4%, a post-pandemic low, but is still much too high, and the reported rate understates the amount of labor market slack.

In fact, given that the FERU was well above its pre-pandemic level, the reported unemployment rate overstated the amount of labor market slack: the unemployment gap $u - u^*$ was much lower than it would have been before the pandemic with the same unemployment rate u.

The mistake was to not look at labor market tightness, which showed a labor market at full employment by mid-2021. Typically, the FERU is utterly stable, so observing unemployment or tightness yields the same insights. But the pandemic was different: the dramatic shift in the Beveridge curve affected the FERU unusually, so simply tracking the unemployment rate was misleading. The insight here is that the efficient tightness is unaffected by shifts in the Beveridge curve, so tightness is much more informative about the state of the labor market when the Beveridge curve is moving. Of course, tracking the unemployment gap (not the unemployment rate) provides the same insights as tracking labor market tightness.

A subtle secondary mistake was to take into account the reduced labor force participation at the time into the calculations. Powell wrote in the quote above that the unemployment rate "understated" the amount of slack. In fact, Powell (2021, chart 2) also displayed an unemployment rate adjusted "for diminished labor force participation induced by the pandemic (as estimated by Federal Reserve Board staff)." That unemployment rate was higher than the regular unemployment rate, prompting the Fed to argue that the unemployment gap was larger than one might think. But as we saw in chapter 9, this reasoning is not correct: even when labor force participation is endogenous, the efficient tightness and unemployment rate remain given by the same formula. So the FERU formula $u^* = \sqrt{uv}$ is also valid when people adjust their labor force participation due to the increased risk of working caused by the coronavirus or to the increased value of staying at home caused by school closures. The Fed should not have used an adjusted unemployment rate in their analysis.

Finally, in 2021:Q2, inflation was elevated, which was another reason to tighten monetary policy. But the Fed might have thought that the inflation spike was temporary, so it might have been reluctant to tighten monetary policy on account of inflation. Powell (2021, p. 5) wrote at the time that:

> Over the 12 months through July, measures of headline and core personal consumption expenditures inflation have run at 4.2% and 3.6%, respectively—well above our 2% longer-run objective.... Inflation at these levels is, of course, a cause for concern. But that concern is tempered by a number of factors that suggest that these elevated readings are likely to prove temporary.

16.7. Application to the slackish business cycle model

The aim of the sufficient-statistic approach is to cover a broad class of models. In this final section, we apply the sufficient-statistic to the slackish business cycle model of chapter 13.

Clearly, the slackish business cycle model with fixed inflation developed in chapter 13 falls into the class of models studied in this chapter. First, social welfare is solely determined by the unemployment rate, so social welfare is maximized when the rate of unemployment is efficient. Second, monetary policy determines the unemployment rate by modulating aggregate demand, without creating any distortions. Hence, the optimal monetary policy is to set the nominal interest rate so as to reach the efficient unemployment rate u^* and eliminate the unemployment gap. Formally, the optimal nominal interest rate i^* must be set so that $u(\theta(i^*)) = u^*$, where the unemployment rate is a function $u(\theta)$ of tightness given by (8.8), and tightness is a function of the interest rate given by (13.12).

Figure 16.3A illustrates the optimal response of monetary policy when the unemployment gap is small. Initially, aggregate demand is given by the dashed aggregate demand curve, and tightness is θ. Since $\theta < \theta^*$, the initial unemployment gap is positive. It is optimal for monetary policy to eliminate the unemployment gap, so the interest rate should drop to stimulate aggregate demand and bring tightness to its efficient level, θ^*.

Since the efficient allocation is in the gray cone in figure 16.3A, optimal monetary policy can be implemented. The gray cone represents the range of aggregate demand curves for any positive interest rate below the current interest rate; the outward boundary of the cone is the highest aggregate demand curve, obtained when the nominal interest rate is 0.

Due to the zero lower bound, the optimal policy is only feasible if $i^* \geq 0$. In figure 16.3B, aggregate demand is initially more depressed, and the efficient allocation falls outside of the gray cone. Hence, monetary policy cannot eliminate the unemployment gap before reaching the zero lower bound. The best that monetary policy can do is reduce the nominal interest rate to 0, and bring the economy to the zero lower bound. There tightness is $\theta^z < \theta^*$, so the unemployment gap remains positive.

Finally, we come back to the conventional situation illustrated in figure 16.3A. There the drop in interest rate to bring tightness up from θ to θ^*, and the unemployment rate down from u to u^*, is given by the sufficient-statistic formula (16.2).

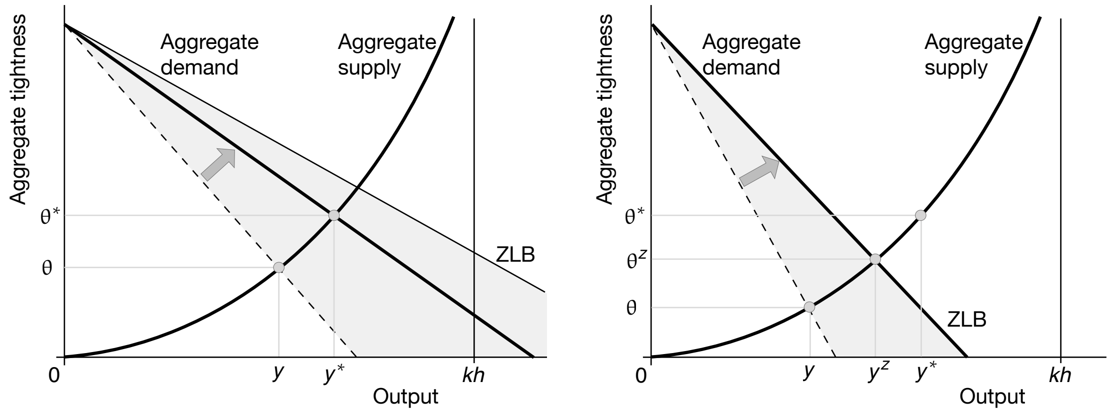
We saw in chapter 13 that the slackish business cycle model can produce the sort of fluctuations in the unemployment gap observed in the US labor market. One naturally wonders whether the slackish business cycle model is also able to match the values of the monetary multiplier estimated in US data—to rationalize empirical estimates of the effectiveness of monetary policy.

To assess whether the model produces a realistic monetary multiplier, we use formula (13.16), which gives the monetary multiplier in the model. At full employment, the multiplier equals

$$\frac{du}{di} = \frac{1 - u^*}{\delta - r^*}.$$

To assess the value of the monetary multiplier, we must calibrate u^*, δ, and r^*. For the FERU, we adopt the average US value between 1948 and 2019, which is $u^* = 4.2\%$ (chapter 12). For the natural real interest rate and time discount rate, we adopt the calibration from chapter 13: $r^* = 0.5\%$ and $\delta = 91\%$. From these estimates, we calibrate the monetary multiplier to be $du/di = (1 - 0.042)/(0.91 - 0.005) = 1.06$.

The calibrated value of the monetary multiplier (1.06) is higher than the estimate from



A. Small initial unemployment gap: optimal monetary policy restores efficient unemployment

B. Large initial unemployment gap: optimal monetary policy reaches the ZLB

FIGURE 16.3. Optimal monetary policy in the slackish business cycle model

The aggregate supply and aggregate demand curves are constructed in figure 13.1B. The ZLB curve is constructed in figure 13.4A. The tightness θ^* and output y^* constitute the efficient allocation.

Coibion (2012)'s hybrid approach (0.5). But it is quite close to the estimate from Coibion (2012)'s narrative approach (0.93). The takeaway is that the monetary multiplier in the basic slackish model is fairly consistent with the estimates coming from the narrative approach, but is much higher than the estimates coming from VAR approaches. If we used a lower value of the time discount factor in our calibration, closer to the median estimate, the calibrated value of the monetary multiplier would be higher, and further away from empirical estimates. So it seems that the slackish business cycle model, in its basic form, predicts effects of monetary policy that are stronger than what we observe empirically.

16.8. Summary

In this chapter, we have seen that if inflation is fixed or the inflation-slack coincidence holds, the optimal monetary policy is to adjust interest rates to eliminate the slack gap entirely. The central bank should lower rates in bad times, when slack is inefficiently high, and raise rates in good times, when slack is inefficiently low.

In fact, optimal monetary policy is given by a simple formula that relates the optimal interest rate to two sufficient statistics: the slack gap and the monetary multiplier (the effect of the federal funds rate on the slack rate). In the United States, the monetary multiplier is about 0.5: the unemployment rate drops by 0.5pp when the nominal interest rate is cut by 1pp. The formula therefore indicates that the Fed should lower the federal funds rate by 2 percentage points for any 1 percentage point of unemployment gap.

When inflation responds to monetary policy and the inflation-slack coincidence fails, optimal monetary policy is more complex but still calls for reducing the slack gap. In this case, it is optimal for the Fed to tolerate inefficiently high slack when inflation is above target, and to tolerate inefficiently low slack when inflation is below target. The sizes of the deviations of slack and inflation from their targets are determined by the welfare cost of inflation relative to slack and by the slope of the Phillips curve.

Finally, we examine why the US economy has generally been too slack in the past century. We find that the price-stability mandate is not the answer: most of the time, when unemployment was too high, inflation was too low. Other constraints on monetary policy, such as political pressure and the zero lower bound, as well as misestimates of full employment, are more plausible explanations.

Bibliography

Bernanke, Ben S. 2022. *21st Century Monetary Policy*. New York: W. W. Norton.

Bernanke, Ben S. and Alan S. Blinder. 1992. "The Federal Funds Rate and the Channels of Monetary Transmission." *American Economic Review* 82 (4): 901–921. https://www.jstor.org/stable/2117350.

Bernanke, Ben S., Jean Boivin, and Piotr Eliasz. 2005. "Measuring the Effects of Monetary Policy: A Factor-Augmented Vector Autoregressive (FAVAR) Approach." *Quarterly Journal of Economics* 120 (1): 387–422. https://doi.org/10.1162/0033553053327452.

BLS. 2025a. "Consumer Price Index for All Urban Consumers: All Items in US City Average." FRED, Federal Reserve Bank of St. Louis. https://fred.stlouisfed.org/series/CPIAUCNS.

BLS. 2025b. "Consumer Price Index for All Urban Consumers: All Items Less Food and Energy in US City Average." FRED, Federal Reserve Bank of St. Louis. https://fred.stlouisfed.org/series/CPILFESL.

Board of Governors of the Federal Reserve System. 2012. "Statement on Longer-Run Goals and Monetary Policy Strategy." https://perma.cc/K4GA-UJNC.

Board of Governors of the Federal Reserve System. 2025. "Federal Funds Effective Rate." FRED, Federal Reserve Bank of St. Louis. https://fred.stlouisfed.org/series/FEDFUNDS.

CBO. 2021. "Natural Rate of Unemployment (Short-Term)." FRED, Federal Reserve Bank of St. Louis. https://fred.stlouisfed.org/series/NROUST.

Christiano, Lawrence J., Martin Eichenbaum, and Charles Evans. 1996. "The Effects of Monetary Policy Shocks: Evidence from the Flow of Funds." *Review of Economics and Statistics* 78 (1): 16–34. https://doi.org/10.2307/2109845.

Christiano, Lawrence J., Martin Eichenbaum, and Charles L. Evans. 1999. "Monetary Policy Shocks: What Have We Learned and to What End?" In *Handbook of Macroeconomics,* vol. 1, edited by John B. Taylor and Michael Woodford, chap. 2. Amsterdam: Elsevier. https://doi.org/10.1016/S1574-0048(99)01005-8.

Coibion, Olivier. 2012. "Are the Effects of Monetary Policy Shocks Big or Small?" *American Economic Journal: Macroeconomics* 4 (2): 1–32. https://doi.org/10.1257/mac.4.2.1.

Department of Veteran Affairs. 2023. "America's Wars." https://perma.cc/GYQ7-DHYL.

Eichengreen, Barry and Peter Temin. 2000. "The Gold Standard and the Great Depression." *Contemporary European History* 9 (2): 183–207. https://doi.org/10.1017/

S0960777300002010.

Friedman, Milton and Anna Schwartz. 1963. *A Monetary History of the United States, 1867–1960*. Princeton, NJ: Princeton University Press.

Greenlaw, David, James D. Hamilton, Ethan Harris, and Kenneth D. West. 2018. "A Skeptical View of the Impact of the Fed's Balance Sheet." NBER Working Paper 24687. https://doi.org/10.3386/w24687.

Hess, Gregory D. and Cameron A. Shelton. 2016. "Congress and the Federal Reserve." *Journal of Money, Credit and Banking* 48 (4): 603–633. https://doi.org/10.1111/jmcb.12312.

Kaya, Ayse, Stephen Golub, Mark Kuperberg, and Feng Lin. 2019. "The Federal Reserve's Dual Mandate and the Inflation-Unemployment Tradeoff." *Contemporary Economic Policy* 37 (4): 641–651. https://doi.org/10.1111/coep.12422.

Kuttner, Kenneth N. 2018. "Outside the Box: Unconventional Monetary Policy in the Great Recession and Beyond." *Journal of Economic Perspectives* 32 (4): 121–146. https://doi.org/10.1257/jep.32.4.121.

Michaillat, Pascal and Emmanuel Saez. 2021. "Resolving New Keynesian Anomalies with Wealth in the Utility Function." *Review of Economics and Statistics* 103 (2): 197–215. https://doi.org/10.1162/rest_a_00893.

NBER. 2023. "US Business Cycle Expansions and Contractions." https://perma.cc/B4LH-BVT8.

Powell, Jerome H. 2021. "Opening Remarks: Monetary Policy in the Time of COVID." In *Proceedings of Jackson Hole Economic Policy Symposium*, pp. 1–15. Kansas City, MO: Federal Reserve Bank of Kansas City. https://perma.cc/MY7X-VQTJ.

Ramey, Valerie A. 2016. "Macroeconomic Shocks and Their Propagation." In *Handbook of Macroeconomics*, vol. 2A, edited by John B. Taylor and Harald Uhlig, chap. 2. Amsterdam: North-Holland. https://doi.org/10.1016/bs.hesmac.2016.03.003.

Ramey, Valerie A. and Matthew D. Shapiro. 1998. "Costly Capital Reallocation and the Effects of Government Spending." *Carnegie-Rochester Conference Series on Public Policy* 48: 145–194. https://doi.org/10.1016/S0167-2231(98)00020-7.

Romer, Christina D. 2021. "The Fiscal Policy Response to the Pandemic." *Brookings Papers on Economic Activity* 52 (1): 89–110. https://doi.org/10.1353/eca.2021.0009.

Romer, Christina D. and David Romer. 2004. "A New Measure of Monetary Policy Shocks: Derivation and Implications." *American Economic Review* 94 (4): 1055–1084. https://doi.org/10.1257/0002828042002651.

Romer, David and Christina Romer. 1989. "Does Monetary Policy Matter? A New Test in the Spirit of Friedman and Schwartz." *NBER Macroeconomics Annual* 4: 121–184. https://doi.org/10.1086/654103.

Shapiro, Adam and Daniel J. Wilson. 2019. "The Evolution of the FOMC's Explicit Inflation Target." FRBSF Economic Letter 2019-12. https://perma.cc/TU8A-EPLR.

Svensson, Lars E. O. 2010. "Inflation Targeting." In *Handbook of Monetary Economics*, vol. 3B, edited by Benjamin M. Friedman and Michael Woodford, chap. 22. Elsevier. https://doi.org/10.1016/B978-0-444-53454-5.00010-4.

Taylor, John B. 1993. "Discretion versus Policy Rules in Practice." *Carnegie-Rochester Conference Series on Public Policy* 39: 195–214. https://doi.org/10.1016/0167-2231(93)90009-L.

Taylor, John B. and John C. Williams. 2010. "Simple and Robust Rules for Monetary Policy." In *Handbook of Monetary Economics*, vol. 3B, edited by Benjamin M. Friedman and Michael Woodford, chap. 15. Amsterdam: Elsevier. https://doi.org/10.1016/B978-0-444-53454-5.00003-7.

Thornton, Daniel L. 2011. "What Does the Change in the FOMC's Statement of Objectives Mean?" Federal Reserve Bank of St. Louis Economic Synopses 1. https://perma.cc/28JH-DRG8.

Weir, Margaret. 1987. "Full Employment as a Political Issue in the United States." *Social Research* 54 (2): 377–402. https://www.jstor.org/stable/40970462.